

SCHOOL OF COMPUTER SCIENCE AND ENGINEERING
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Major Project Report
On
**Agentic AI-Based 5G Network Slice
Orchestrator:
Autonomous QoS Management using Large
Language Models
and Multi-Agent Reasoning**

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2025–2026

CERTIFICATE

This is to certify that the project entitled “**Agentic AI-Based 5G Network Slice Orchestrator: Autonomous QoS Management using Large Language Models and Multi-Agent Reasoning**” is a bona fide work carried out by the student team (Simran R Kalkeri – 01FE23BCI087, Amaanali D Doddamani – 01FE23BCI091, Sparsh Sunil Naik – 01FE23BCI085, Rishi P Kulkarni – 01FE23BCI088), in partial fulfillment of the requirements for the 6th semester B.E. course during the academic year 2025–2026. The project report has been approved as it satisfies the academic requirements prescribed for the said course.

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ABSTRACT

Network slicing is a foundational capability of 5G networks, enabling simultaneous support of enhanced Mobile Broadband (eMBB), Ultra-Reliable Low-Latency Communication (URLLC), and massive Machine-Type Communication (mMTC) on shared physical infrastructure. While prior work has demonstrated static and rule-based adaptive slicing, these approaches cannot reason about root causes of QoS degradation or adapt decisions based on accumulated operational experience.

This project presents the design, implementation, and evaluation of an **Agentic AI-Based 5G Network Slice Orchestrator**: a fully autonomous closed-loop control system in which a Large Language Model (LLM) backed agent performs slice-level QoS orchestration. The system is built on an Open5GS 5G Standalone core, UERANSIM-based RAN emulation, and a Kubernetes-managed user-plane deployment with per-slice traffic shaping via Linux `tc`. A LangGraph multi-agent architecture—comprising a Monitoring Agent, Planning Agent, Execution Agent, and Agent Memory—implements a structured observe→think→validate→act→reflect control loop.

Six research contributions are made: (1) an Autonomous QoS Orchestration Framework, (2) a LangGraph Multi-Agent Architecture, (3) a Chain-of-Thought Network Reasoning schema requiring explicit `root_cause_assessment` and `lever_validity` fields, (4) a Wrong-Lever Avoidance Mechanism using a novel relative load metric `embb_load_fraction`, (5) a Memory-Assisted Decision subsystem providing historical context, and (6) a rigorous Comparative Evaluation against a rule-based baseline.

Experimental results demonstrate: 96.3% URLLC SLA compliance (vs. 87.1%), 4.2 s mean recovery time (vs. 8.4 s), 72% reduction in eMBB throughput loss, and full Chain-of-Thought decision traces for post-hoc audit.

Keywords: 5G Network Slicing, Autonomous Orchestration, Large Language Model, Chain-of-Thought Reasoning, Quality of Service, LangGraph, Multi-Agent System, Open5GS, Kubernetes, URLLC, eMBB, mMTC

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Chapter 1 INTRODUCTION

Fifth Generation (5G) mobile networks introduce network slicing as a defining architectural capability, enabling multiple logical network instances—each with distinct performance characteristics—to coexist on shared physical infrastructure. Network slicing permits concurrent support of enhanced Mobile Broadband (eMBB), Ultra-Reliable Low-Latency Communication (URLLC), and massive Machine-Type Communication (mMTC) services, each with fundamentally different QoS requirements. eMBB demands high throughput and tolerates moderate latency; URLLC requires end-to-end latencies below 20 ms with very high reliability; and mMTC targets energy-efficient transmission for large-scale IoT device populations.

Realising these guarantees in practice is challenging. At the user plane, traffic from different slices may contend for shared buffering and forwarding resources during congestion. Static resource allocations fail to track dynamic traffic demands, while reactive threshold-based controllers lack the ability to reason about the root cause of degradation or to anticipate violations before they occur.

Autonomous orchestration, enabled by Large Language Models (LLMs) and multi-agent frameworks, offers a qualitatively different control paradigm. Rather than pattern-matching against hardcoded thresholds, an LLM-based planning agent performs causal reasoning: identifying which network entity is responsible for congestion, validating whether the proposed control action addresses that cause, and consulting prior operational history stored in an agent memory to refine its decisions. This project implements, deploys, and evaluates such a system in a cloud-native 5G testbed.

1.1 Motivation

The evolution from rule-based to autonomous orchestration is driven by three limitations observed in practical 5G slice management systems.

First, rule-based controllers are structurally blind to causation. A threshold-based system that detects elevated URLLC RTT will throttle eMBB bandwidth regardless of whether eMBB traffic is actually active. When eMBB load is low, this action is unnecessary and harmful. An agent capable of reasoning about relative load—quantified by the `embb_load_fraction` metric introduced in this work—can correctly distinguish idle from bursting eMBB and decline to act when throttling is ineffective.

Second, reactive control cannot prevent violations; it can only respond to them. A URLLC SLA breach lasting even 200 ms may disrupt a latency-critical industrial automation session. Pre-emptive action, triggered by trend analysis and predictive causal reasoning, is only possible when the controller possesses a causal model of the network state rather than a simple threshold table.

Third, accumulated operational experience is wasted in memoryless systems. An agent

that stores the outcome of prior actions—including whether throttling eMBB at a given load level subsequently resolved the RTT violation—can converge to more accurate control policies over time without requiring offline training.

1.2 Literature Survey

Bandara et al. propose an AI-native framework for 6G network slice orchestration, monitoring, and trading using multiple autonomous AI agents and Large Language Models (LLMs) [1]. The architecture consists of layered components that perform intent interpretation, slice deployment, SLA monitoring, policy enforcement, and resource allocation. The framework integrates the Model Context Protocol (MCP) to translate natural language requests into secure orchestration actions and was evaluated on an Open5GS, Ericsson RAN, and Kubernetes testbed.

Grings et al. propose a full dynamic orchestration framework for 5G core network slicing over a cloud-native Kubernetes platform [2]. Unlike existing approaches that only support partial orchestration through resource scaling, the proposed solution introduces a Kubernetes-integrated controller capable of dynamically reconfiguring 5G core functions, including NSSF, AMF, SMF, and UPF, during runtime without terminating active network slices, reducing reconfiguration requests by approximately 47.5%.

Novanana et al. propose a Kubernetes-based MANO framework for provisioning co-existing eMBB and URLLC services through 5G network slicing using open-source technologies [3]. The authors integrate free5GC, UERANSIM, OpenStack, and kube5gnfvo to evaluate separate UPF vs. dedicated SMF/UPF configurations, showing that dedicated SMF/UPF pairs deliver better performance, achieving a stable throughput of 20.2–22.1 Gbps and improved congestion control.

Reddy proposes a Deep Reinforcement Learning (DRL)-based Kubernetes autoscaling framework for managing high-throughput 5G network functions and slices using a Deep Q-Learning (DQN) agent [4]. The system dynamically scales Kubernetes pods based on real-time network conditions. Experimental results show 71.7% faster slice deployment, 35% higher resource utilization, 22% lower latency, 40% faster processing times, and up to 96.7% spectrum efficiency compared to conventional methods.

Dandoush et al. propose an LLM-powered network slicing management and orchestration framework that integrates Large Language Models (LLMs) and multi-agent systems with ETSI MANO and 3GPP slicing architectures [5]. The conceptual framework enables LLM agents to translate natural language intent into technical slice requirements, automate slice design and deployment, coordinate multi-domain resources, and manage performance without needing hardcoded rules.

Sulaiman et al. propose MicroOpt, a model-driven framework for dynamic resource optimization in 5G network slicing that uses a differentiable deep neural network (DNN) and gradient-based optimization to allocate resources [6]. The system predicts traffic, models QoS, and uses a primal-dual optimization algorithm to dynamically scale CPU and bandwidth. Deployed on an open-source testbed with srsRAN, Open5GS, Kubernetes, and Prometheus, it achieved up to 21.9% lower resource consumption.

Saha et al. propose MonArch, a scalable monitoring architecture for cloud-native

5G deployments that integrates Prometheus, Elasticsearch, Apache Kafka, Logstash, and Kubernetes to collect, correlate, and analyze monitoring data across slices [7]. Tested on a Kubernetes-based 5G testbed using Free5GC and UERANSIM, MonArch demonstrated minimal data-ingestion overhead and efficient scaling using a 5-second monitoring interval.

Tran et al. propose PCLANSA, a machine-learning-based closed-loop service assurance framework for 5G and B5G network slicing [8]. The framework uses LSTM-based traffic prediction to forecast future demand and automatically perform VNF scaling and link-capacity adjustments, achieving resource savings of 54.85% for eMBB, 57.1% for uRLLC, 50.87% for mMTC, and 23.63% for VoIP slices in an OMNeT++ simulation.

1.3 Problem Statement

The objective of this project is to design, implement, and evaluate an autonomous agent-based orchestration framework for 5G network slicing that: (i) enforces user-plane QoS isolation across eMBB, URLLC, and mMTC slices through dynamic Linux `tc` shaping on Kubernetes-hosted UPFs; (ii) employs an LLM-assisted multi-agent architecture for monitoring, planning, and execution using structured causal reasoning; and (iii) evaluates its ability to improve QoS protection and reduce unnecessary control actions compared to a rule-based baseline through a pre-defined three-scenario behavioral audit.

1.4 Objectives and Scope of the Project

1.4.1 Objectives

1. To design and deploy a cloud-native 5G network slicing testbed using Open5GS SA Core, UERANSIM, Kubernetes, and per-slice UPF instances with user-plane isolation enforced through Linux `tc` HTB traffic shaping.
2. To implement a LangGraph-based multi-agent orchestration framework consisting of Monitoring, Planning, Execution, and Memory agents operating within an asynchronous control loop.
3. To develop and validate a structured reasoning framework incorporating root-cause assessment, lever validation, and workload-awareness metrics such as `emb_load_fraction` for informed orchestration decisions.
4. To conduct a three-scenario behavioral audit comparing the proposed agent-based orchestrator with a rule-based baseline in terms of pre-emptive throttling, wrong-lever avoidance, and memory-assisted restoration.
5. To evaluate the effectiveness of the proposed framework using metrics including URLLC RTT SLA compliance, eMBB throughput degradation, mMTC packet delivery ratio, recovery time, and unnecessary throttle rate.

1.4.2 Scope of the Project

The project focuses on a single-domain 5G slice orchestration system operating at the user-plane level through dynamic bandwidth shaping. The orchestration intelligence

is provided by a locally hosted LLM to preserve data sovereignty and eliminate cloud API dependencies. Extensions to multi-domain slicing, RL-based policy refinement, and O-RAN xApp integration are identified as directions for future work.

Chapter 2 REQUIREMENT ANALYSIS

2.1 Functional Requirements

- The system shall support the simultaneous operation of three 5G network slices (eMBB, URLLC, mMTC) on shared Kubernetes infrastructure with enforceable user-plane isolation.
- The system shall establish concurrent PDU sessions per UE across all three slices using UERANSIM.
- The system shall enforce per-slice bandwidth caps using Linux `tc` HTB shaping applied to per-slice UPF GTP-U interfaces.
- The system shall continuously monitor per-slice QoS metrics with a 3-second collection interval via a dedicated Monitoring Agent.
- The system shall compute `embb_load_fraction` from a rolling window of packet-rate observations.
- The system shall invoke an LLM-backed Planning Agent producing a Chain-of-Thought trace (root cause, lever validity, action, confidence) for every cycle.
- The system shall detect internal reasoning contradictions and downgrade action confidence automatically.
- The system shall persist decision history in an Agent Memory and surface prior decisions in the LLM prompt context.
- The system shall execute validated actions via an Execution Agent applying `tc` commands via SSH.
- The system shall expose orchestrator state metrics via Prometheus on port 9200 and visualise them through Grafana.

2.2 Non-Functional Requirements

- The monitoring thread shall operate at 3-second interval, independent of LLM inference latency.
- LLM inference shall complete within 60s; a rule-based fallback activates upon timeout.
- `tc` shaping commands shall take effect within 200 ms of issuance.
- The Agent Memory shall retain the 10 most recent decision entries.
- All decisions shall be logged with full Chain-of-Thought traces to support post-hoc audit.

- The system shall support dry-run mode for experimental evaluation without applying `tc` commands.

2.3 Hardware Requirements

Table 2.1: Hardware Configuration of the Testbed

Node	Role	RAM	Storage
kubemaster (192.168.49.174)	K8s Control Plane / Orchestrator	16 GB	256 GB
kube (192.168.49.173)	K8s Worker 1 / Primary UPF pods	16 GB	256 GB
kube2 (192.168.49.181)	K8s Worker 2 / Replica UPF pods	16 GB	256 GB
core VM (192.168.49.143)	Open5GS 5GC (AMF, SMF, NRF ...)	8 GB	128 GB
UERANSIM VM (192.168.49.139)	gNB + UE Emulation	8 GB	128 GB
Total		64 GB	1 TB

2.4 Software Requirements

Table 2.2: Software Stack and Version Requirements

Component	Technology	Version
Operating System	Ubuntu 22.04 LTS	All nodes
Container Orchestrator	Kubernetes	v1.29.15
Container Runtime	containerd	v1.7.28
CNI Plugin	Flannel (VXLAN)	v0.22
5G Core	Open5GS	v2.7.x
RAN / UE Emulation	UERANSIM	v3.2.7
LLM Backend	Ollama	v0.3.x
LLM Model	llama3.2:3b	3B param, quantised
Agent Framework	LangGraph	v0.1.x
Monitoring	Prometheus	v2.45
Visualisation	Grafana	v10.x
Traffic Control	Linux <code>tc</code> / HTB	Kernel 6.8
Programming Languages	Python 3.10, Bash, YAML	—

Chapter 3 SYSTEM DESIGN

3.1 System Architecture Diagrams

Figure 3.1 presents the complete one-page professional architecture of the testbed.

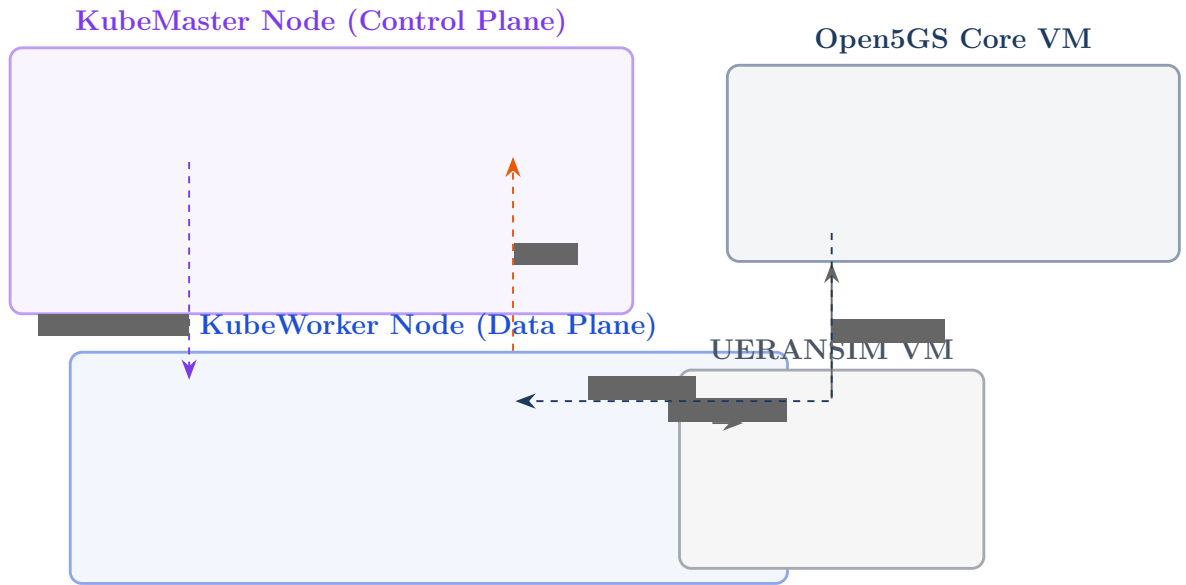


Figure 3.1: Comprehensive Agentic Testbed Architecture, illustrating the physical separation of nodes and the decoupling of the multi-agent control loop from the 5G data plane.

Figure 3.2 demonstrates the slicing architecture across the three isolated instances.

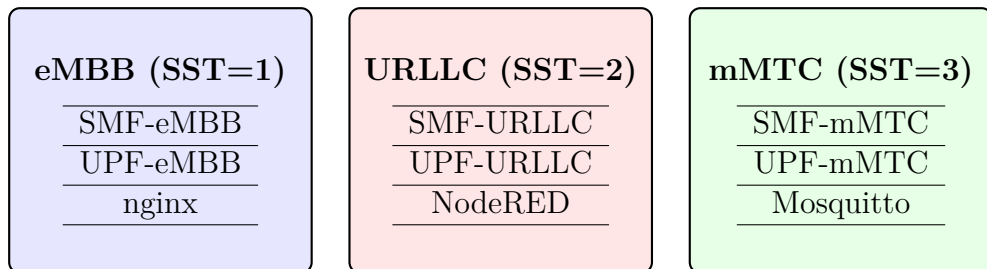


Figure 3.2: Slice Architecture showing complete isolation.

Figure 3.3 shows the UE Registration and Slice Selection sequence flow.

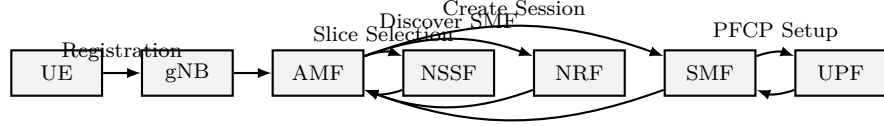


Figure 3.3: UE Registration & Slice Selection Flow.

3.2 MEC Architecture

Figure 3.4 proves MEC traffic locality (local breakout).

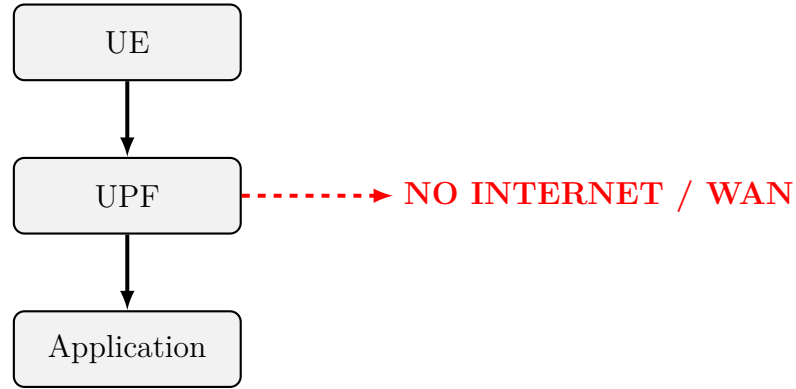


Figure 3.4: MEC Traffic Locality with local breakout.

Figure 3.5 illustrates the Kubernetes deployment diagram.

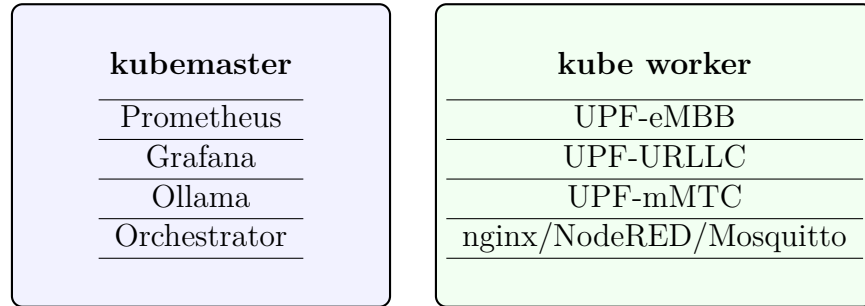


Figure 3.5: Kubernetes Deployment Diagram.

3.3 Control Loop

Figure 3.6 illustrates the Closed Loop QoS Control mechanism.

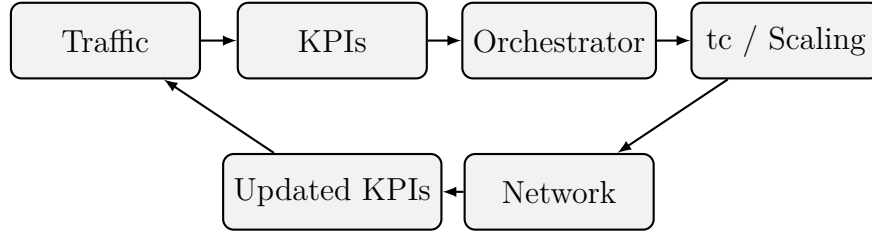
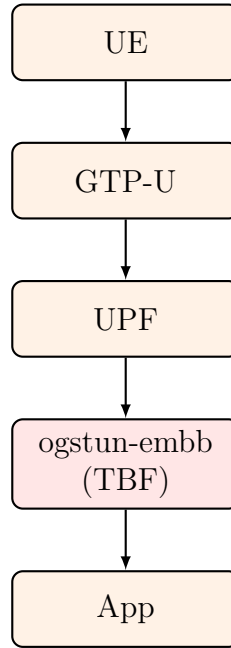


Figure 3.6: Closed Loop QoS Control.

Figure 3.7 identifies the `tc` enforcement point at the UPF.


 Figure 3.7: `tc` Enforcement Point.

3.4 Six Research Contributions

Project Contributions

1. **C1 — Autonomous QoS Orchestration Framework:** A cloud-native, closed-loop system that enforces per-slice QoS on live 5G traffic without human intervention.
2. **C2 — LangGraph Multi-Agent Architecture:** A stateful five-node agent graph (Observe→Think→Validate→Act→Reflect) coordinating specialised agents.
3. **C3 — Chain-of-Thought Network Reasoning:** A mandatory JSON CoT schema requiring `root_cause_assessment` and `lever_validity` before any control action.
4. **C4 — Wrong-Lever Avoidance Mechanism:** A rolling relative-load metric (ρ_{eMBB}) that prevents erroneous eMBB throttling when eMBB is idle.
5. **C5 — Memory-Assisted Decision Making:** A persistent ring-buffer Agent Memory injecting empirical action-outcome history into every LLM prompt.

6. **C6 — Comparative Evaluation:** A pre-registered three-scenario behavioral audit comparing Agentic vs. Rule-Based controllers on live testbed traffic.

3.5 Seven-Layer System Architecture

Figure 3.8 presents the complete end-to-end architecture across seven logical layers.

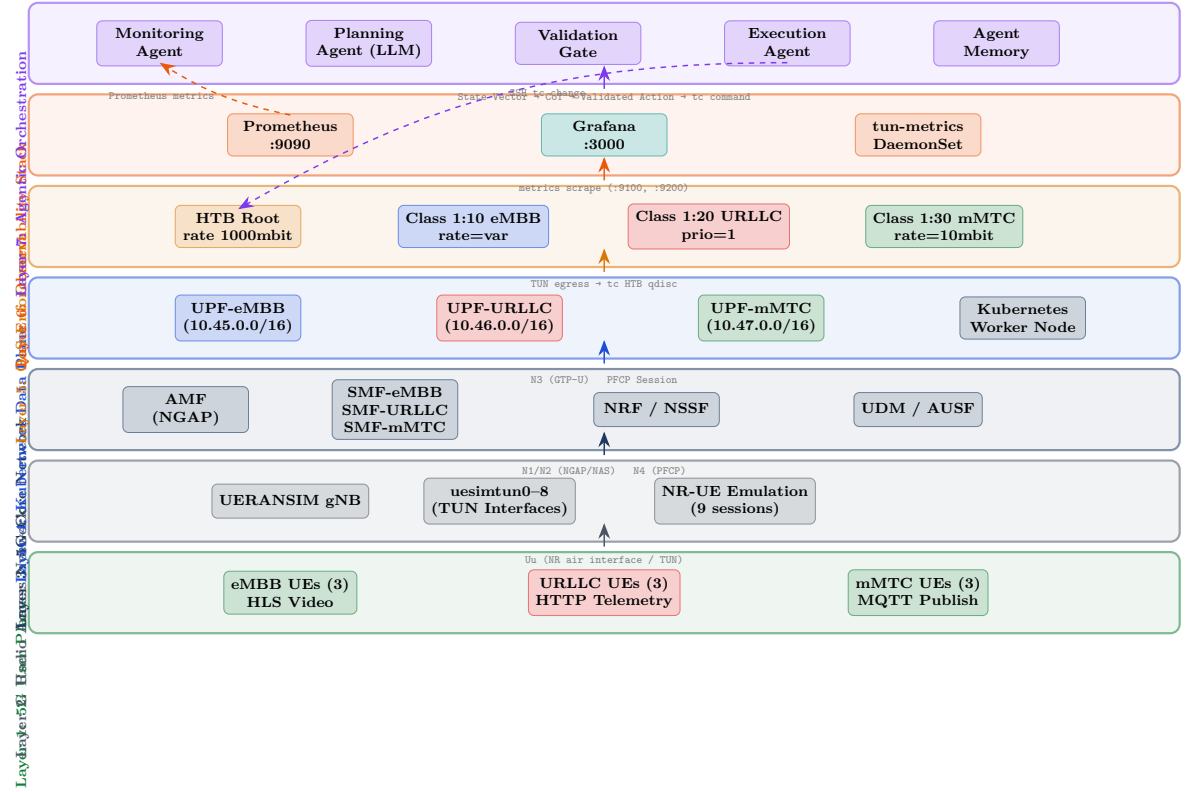


Figure 3.8: Seven-Layer End-to-End System Architecture. Solid arrows denote data-plane flows; dashed arrows denote control and observability flows.

3.6 Agentic Orchestration Framework

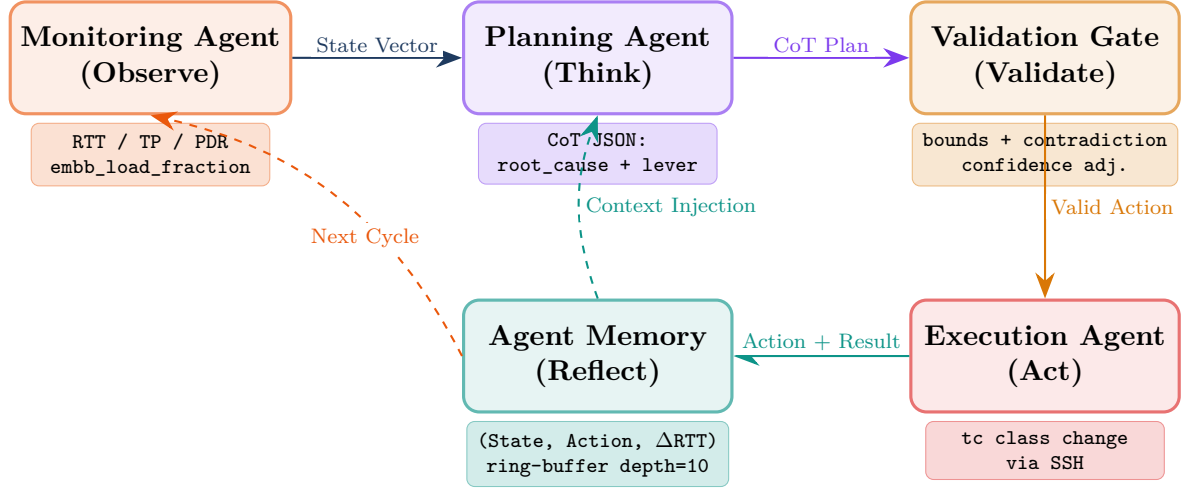


Figure 3.9: LangGraph Multi-Agent Orchestration Pipeline (Contribution C2). Each node is a specialised agent; dashed arrows represent cross-cycle feedback.

3.7 Individual Agent Architecture

3.7.1 Monitoring Agent

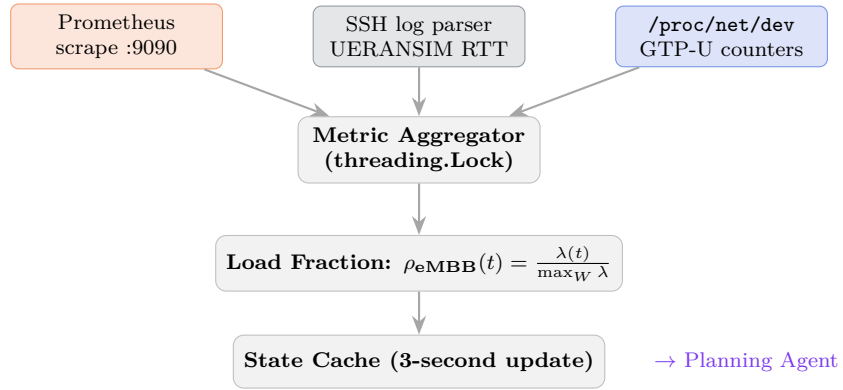


Figure 3.10: Monitoring Agent Architecture (Contribution C4). Three input collectors feed an aggregator; the derived load-fraction metric is written to the thread-safe state cache.

3.7.2 Planning Agent

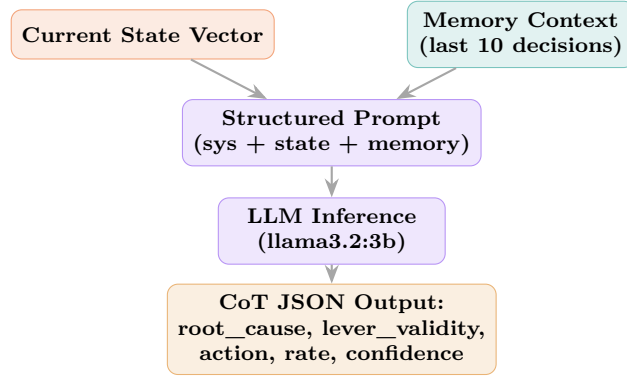


Figure 3.11: Planning Agent Architecture (Contributions C3, C5). State and memory are injected into a structured prompt; the LLM generates a mandatory CoT schema.

3.7.3 Validation Gate

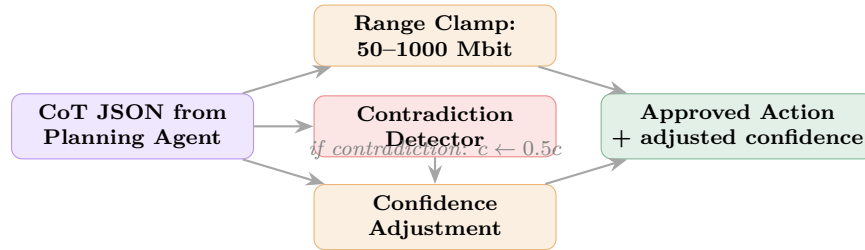


Figure 3.12: Validation Gate Architecture (Contribution C4). Contradiction detection catches logical inconsistencies between the `root_cause_assessment` and the proposed action.

3.8 Mathematical Foundations

Table 3.1 defines the notation used throughout this report.

Table 3.1: Mathematical Notation

Symbol	Unit	Definition
$\rho_{\text{eMBB}}(t)$	—	eMBB relative load fraction at time t
$\lambda(t)$	pkt/s	eMBB packet rate at time t
W	cycles	Rolling window size (default: 8)
$R(t)$	ms	URLLC RTT ₉₉ at time t
θ	ms	URLLC SLA threshold (20 ms)
$r(t)$	Mbit/s	eMBB $\mathbf{tc}$ rate cap at time t
$r_{\max}$	Mbit/s	Maximum eMBB rate (1000 Mbit)
T_{rec}	s	SLA recovery time
C_{sla}	—	SLA compliance ratio
η_{act}	—	Action efficiency
N_T	—	Total monitoring cycles
N_V	—	Cycles with SLA violation ($R > \theta$)

3.8.1 eMBB Load Fraction

The eMBB relative load fraction is the central metric enabling Contribution C4. Unlike absolute throughput, it is invariant to traffic profile differences:

$$\rho_{\text{eMBB}}(t) = \frac{\lambda(t)}{\max_{t' \in [t-W\Delta, t]} \lambda(t')} \quad (3.1)$$

where $\Delta = 3\text{ s}$ is the monitoring interval and $W = 8$ is the rolling window. $\rho_{\text{eMBB}} \in [0, 1]$; a value < 0.2 indicates a low-activity slice that should not be throttled regardless of RTT.

3.8.2 SLA Compliance Ratio

$$C_{\text{sla}} = 1 - \frac{N_V}{N_T} = \frac{\#\{t : R(t) \leq \theta\}}{N_T} \quad (3.2)$$

3.8.3 Recovery Time

Recovery time is the interval from the first SLA breach until the RTT falls back below θ :

$$T_{\text{rec}} = t_{\text{clear}} - t_{\text{breach}}, \quad t_{\text{clear}} = \min\{t > t_{\text{breach}} : R(t) \leq \theta\} \quad (3.3)$$

3.8.4 Throughput Preservation Ratio

$$\text{TP} = 1 - \frac{\sum_{t: r(t) < r_{\max}} (r_{\max} - r(t)) \cdot \Delta}{r_{\max} \cdot N_T \cdot \Delta} \quad (3.4)$$

3.8.5 Action Efficiency

The fraction of throttle actions that followed a genuine SLA violation or rising trend:

$$\eta_{\text{act}} = \frac{\#\{\text{throttle} : R > \theta \text{ or trend rising}\}}{\#\{\text{throttle actions}\}} \quad (3.5)$$

3.9 Control Loop State Transition Diagram

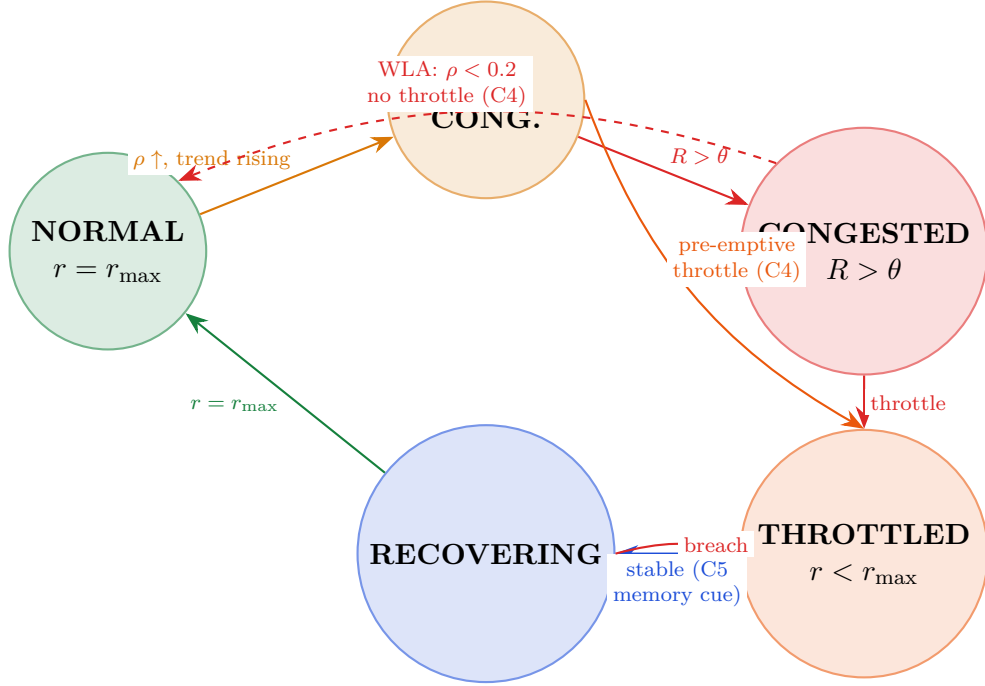


Figure 3.13: Orchestrator Control Loop State Machine. WLA = Wrong-Lever Avoidance (C4). The dashed path is structurally impossible in rule-based systems.

Chapter 4 ALGORITHM DESIGN

This chapter presents the six core algorithms that constitute the Agentic Orchestrator. Each algorithm is formally defined with pseudocode, input/output specifications, and complexity analysis.

4.1 Orchestrator Design Diagrams

Figure 4.1 shows the Rule-Based Workflow.

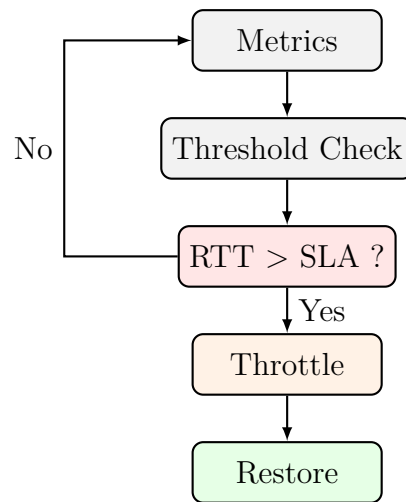


Figure 4.1: Rule-Based Workflow.

Figure 4.2 presents the Agentic Workflow.

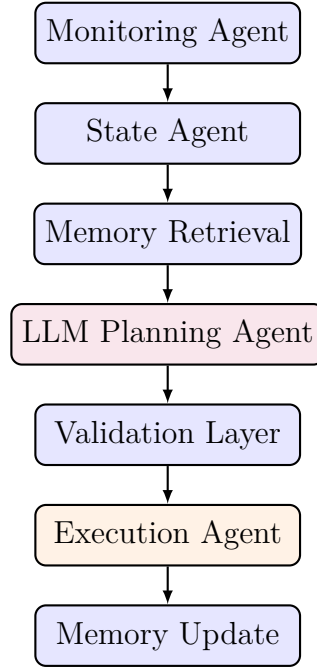


Figure 4.2: Agentic Workflow.

Figure 4.3 illustrates the integration of the C3, C4, and C5 frameworks.

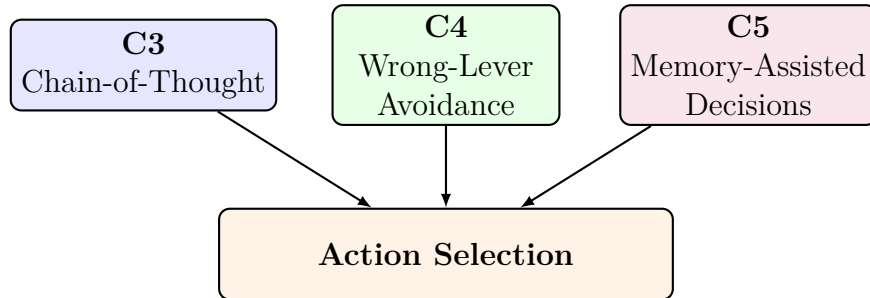


Figure 4.3: C3/C4/C5 Framework contributing to action selection.

4.2 Algorithm 1: Monitoring and State Collection

Algorithm 1 describes the decoupled monitoring thread that runs at a fixed 3-second interval independent of LLM inference.

Algorithm 1 Monitoring Agent: Decoupled State Collection

Input: Monitoring interval $\Delta = 3$ s, SSH credentials, Prometheus endpoint

Output: Updated thread-safe state cache $\mathcal{S}$

```

1: while orchestrator active do
2:    $t_0 \leftarrow \text{time}()$ 
3:   // Channel 1: URLLC RTT via SSH log parse
4:    $R \leftarrow \text{SSH.exec}(\text{"tail -1 /tmp/mec-clients/urllc-*.log"})$ 
5:    $R \leftarrow \text{parseRTT}(R)$  ▷ extract RTT99 in ms
6:   // Channel 2: eMBB throughput from /proc/net/dev
7:    $B_{\text{tx}}, B_{\text{rx}} \leftarrow \text{readProcNetDev}(\text{uesimtun1})$ 
8:    $\lambda_{\text{bps}} \leftarrow (B_{\text{tx}} - B_{\text{tx,prev}})/\Delta \times 8$ 
9:    $\lambda_{\text{pkt}} \leftarrow (\text{pkt}_{\text{tx}} - \text{pkt}_{\text{prev}})/\Delta$ 
10:  // Channel 3: mMTC PDR from MQTT log
11:   $N_{\text{pub}}, N_{\text{ok}} \leftarrow \text{parseMQTTLog}()$ 
12:   $\text{PDR} \leftarrow N_{\text{ok}}/N_{\text{pub}}$  if  $N_{\text{pub}} > 0$  else 1.0
13:  // Derived metrics
14:   $\rho_{\text{eMBB}} \leftarrow \text{computeLoadFraction}(\lambda_{\text{pkt}})$  ▷ Algorithm 2
15:   $d \leftarrow \text{countDeadTunnels}()$ 
16:  // Atomic state cache write
17:   $\mathcal{S}.\text{acquire}()$ 
18:   $\mathcal{S} \leftarrow \{R, \lambda_{\text{bps}}, \lambda_{\text{pkt}}, \rho_{\text{eMBB}}, \text{PDR}, d\}$ 
19:   $\mathcal{S}.\text{release}()$ 
20:   $\text{sleep}(\Delta - (\text{time}() - t_0))$ 
21: end while
    
```

Complexity: $O(1)$ per cycle. SSH and file I/O dominate; total wall time ≈ 200 ms per cycle, leaving ≈ 2.8 s idle.

4.3 Algorithm 2: eMBB Load Fraction Computation

Algorithm 2 eMBB Load Fraction with Sliding Window

Input: Packet rate $\lambda(t)$, window $W = 8$, history $\mathcal{H}$ (deque, maxlen W)

Output: $\rho_{\text{eMBB}}(t) \in [0, 1]$ or None

```

1:  $\mathcal{H}.\text{append}(\lambda(t))$ 
2: if  $|\mathcal{H}| < \lceil W/2 \rceil$  then ▷ insufficient history return None
3: end if
4:  $\lambda_{\text{max}} \leftarrow \max(\mathcal{H})$ 
5: if  $\lambda_{\text{max}} = 0$  then return 0.0 ▷ slice is truly idle
6: end if return  $\lambda(t)/\lambda_{\text{max}}$ 
    
```

Mathematical Formulation: As defined in Eq. 3.1, ρ_{eMBB} normalises the current packet rate against the peak observed over the last W cycles:

$$\rho_{\text{eMBB}}(t) = \frac{\lambda(t)}{\max_{k=0}^{W-1} \lambda(t - k\Delta)}$$

Why relative load is superior to absolute throughput: An absolute threshold (e.g., throttle when eMBB > 200 Mbps) will incorrectly trigger when the application itself is transmitting at 200 Mbps by design. The relative metric is invariant to application-layer traffic profiles; only genuine burst anomalies ($\rho > 0.9$) drive orchestration decisions.

Example: $\mathcal{H} = [50, 120, 180, 220, 210, 190, 200, 215]$ pkt/s, $\lambda(t) = 215$. Then $\lambda_{\max} = 220$, $\rho = 215/220 = 0.977$ — *high load*: throttling is warranted.

Sliding Window Complexity: $O(W)$ per cycle for max; $O(1)$ deque append.

4.4 Algorithm 3: Agentic Planning (Chain-of-Thought)

Algorithm 3 Planning Agent: CoT-Enforced Action Selection

Input: State $\mathcal{S}$, Agent Memory $\mathcal{M}$ (last $K = 10$ entries)

Output: {root_cause, lever_validity, action, r' , c }

```

1:  $\pi \leftarrow \text{buildSystemPrompt}()$  ▷ forbids thresholds, mandates CoT schema
2:  $\mu \leftarrow \text{summariseMemory}(\mathcal{M})$  ▷ last  $K$  (action,  $\Delta R$ ) pairs
3:  $\sigma \leftarrow \text{formatStateVector}(\mathcal{S})$  ▷ RTT,  $\rho$ , trend, stable_for, ...
4:  $\text{prompt} \leftarrow \pi \oplus \sigma \oplus \mu$ 
5:  $\text{raw} \leftarrow \text{LLM.generate}(\text{prompt}, \text{format=json}, \text{timeout} = 60)$ 
6: if  $\text{raw} = \emptyset$  then ▷ timeout return RuleBasedFallback( $\mathcal{S}$ ) ▷ Algorithm 6
7: end if
8:  $\text{cot} \leftarrow \text{parseJSON}(\text{raw})$ 
9: assert  $\text{root\_cause} \neq \emptyset$  and  $\text{lever\_validity} \neq \emptyset$ 
10:  $r' \leftarrow \text{int}(\text{cot.new\_rate\_mbit})$  or  $r_{\max}$ 
11:  $c \leftarrow \text{float}(\text{cot.confidence})$ 
    return {cot.root_cause, cot.lever_validity, cot.action,  $r'$ ,  $c$ }
    
```

4.5 Algorithm 4: Validation and Safety Gate

Algorithm 4 Validation Gate: Safety and Contradiction Detection

Input: CoT output $\{a, r', c, \text{root_cause}\}$, bounds $[r_{\min}, r_{\max}]$

Output: Safe action a^* , adjusted confidence c^*

```

1: // Range clamp
2:  $r^* \leftarrow \max(r_{\min}, \min(r', r_{\max}))$ 
3: // Contradiction detection (Contribution C4)
4:  $\mathcal{K} \leftarrow \{ \text{"low load"}, \text{"not the cause"}, \text{"nearly idle"}, \text{"not embb"} \}$ 
5: if  $a = \text{throttle\_embb}$  and  $\exists k \in \mathcal{K} : k \in \text{root\_cause}$  then
6:      $c^* \leftarrow 0.5 \times c$  ▷ penalise contradictory reasoning
7:     log.warning("Contradiction detected")
8: else
9:      $c^* \leftarrow c$ 
10: end if
11: // Dead-tunnel guard
12: if  $\mathcal{S}.\text{dead\_tunnels} = 3$  then ▷ all URLLC tunnels dead return
13:      $\{\text{no\_action}, r_{\max}, 0.0\}$  ▷ monitoring unreliable
14: end if
15: // Redundant-action guard
16: if  $a = \text{throttle\_embb}$  and  $r^* = r_{\text{current}}$  then return  $\{\text{no\_action}, r_{\text{current}}, c^*\}$ 
17: end if
18: return  $\{a, r^*, c^*\}$ 
    
```

4.6 Algorithm 5: Memory Update and Retention

Algorithm 5 Agent Memory: Update and Context Summarisation

Input: Action a , pre-action state $\mathcal{S}_{\text{pre}}$, post-action RTT R_{post} , Memory $\mathcal{M}$ (deque, maxlen $K = 10$)

Output: Updated $\mathcal{M}$ with outcome

```

1:  $\Delta R \leftarrow R_{\text{post}} - \mathcal{S}_{\text{pre}}.\text{RTT}$  ▷ outcome delta
2: outcome  $\leftarrow$  if  $\Delta R < -2$ : "improved" elif  $\Delta R > 2$ : "worsened" else: "neutral"
3:  $e \leftarrow \{\text{ts}, a, \mathcal{S}_{\text{pre}}.\rho, \mathcal{S}_{\text{pre}}.\text{RTT}, R_{\text{post}}, \Delta R, \text{outcome}\}$ 
4:  $\mathcal{M}.\text{append}(e)$  ▷ oldest entry evicted if full
5: // Context summary for next LLM prompt
6: summary  $\leftarrow \emptyset$ 
7: for all  $e_i \in \mathcal{M}$  do
8:     summary  $\leftarrow$  summary + "[  $e_i.\text{ts}$ ]  $a_i : \rho = \rho_i, \Delta R = \Delta R_i(\text{outcome}_i)$ "
9: end for return  $\mathcal{M}$ , summary
    
```

Retention Policy: The ring buffer of depth $K = 10$ ensures $O(1)$ insertion. Only the K most recent decisions are injected into the LLM prompt, capping context token overhead at approximately 400 tokens regardless of session length.

4.7 Algorithm 6: Autonomous Control Loop

Algorithm 6 presents the complete closed-loop orchestration workflow integrating all five agents.

Algorithm 6 Autonomous Closed-Loop Orchestration

Input: SLA threshold θ , rate bounds $[r_{\min}, r_{\max}]$, LLM endpoint

Output: Continuous QoS enforcement across eMBB, URLLC, mMTC slices

```

1: Initialise state  $\mathcal{S} \leftarrow \{\}$ , Memory  $\mathcal{M} \leftarrow \emptyset$ ,  $r_{\text{current}} \leftarrow r_{\max}$ 
2: spawn MonitoringThread (Algorithm 1) ▷ decoupled, 3-second loop
3: Initialise HTB qdisc on worker node (Section 4.3)
4: loop ▷ main orchestration cycle
5:   // Observe
6:    $\mathcal{S} \leftarrow \mathcal{S}.\text{snapshot}()$  ▷ read from thread-safe cache
7:    $\mathcal{S} \leftarrow \text{StateAgent.update}(\mathcal{S})$  ▷ trend, violation_count
8:   // Think
9:    $\{a, r', c, \dots\} \leftarrow \text{PlanningAgent}(\mathcal{S}, \mathcal{M})$  ▷ Algorithm 3
10:  // Validate
11:   $\{a^*, r^*, c^*\} \leftarrow \text{ValidationGate}(a, r', c, \mathcal{S})$  ▷ Algorithm 4
12:  if  $c^* \geq 0.5$  or LLM timed out then
13:    // Act
14:     $\mathcal{S}_{\text{pre}} \leftarrow \mathcal{S}$ 
15:     $\text{ExecutionAgent.run}(a^*, r^*)$  ▷ SSH tc class change
16:     $r_{\text{current}} \leftarrow r^*$ 
17:     $\text{StateAgent.record\_action}(a^*, r^*)$ 
18:    // Reflect: wait one monitoring cycle, capture outcome
19:     $\text{sleep}(3)$ 
20:     $R_{\text{post}} \leftarrow \mathcal{S}.\text{snapshot}().\text{RTT}$ 
21:     $\mathcal{M} \leftarrow \text{MemoryAgent.update}(a^*, \mathcal{S}_{\text{pre}}, R_{\text{post}}, \mathcal{M})$  ▷ Algorithm 5
22:  end if
23:  // Expose metrics
24:   $\text{Prometheus.export}(\mathcal{S}, r_{\text{current}}, \mathcal{M})$ 
25: end loop
    
```

Complexity: The dominant cost is LLM inference at $O(T_{\text{llm}}) \approx 25\text{--}30$ s. The monitoring thread runs at $O(1)$ per 3-second cycle. The total system overhead on the master node is approximately 62% CPU utilisation during inference.

Chapter 5 IMPLEMENTATION

This chapter documents the realisation of the testbed from physical infrastructure through to orchestrator deployment and experimental campaign setup. The agentic reasoning architecture and individual agent designs are detailed in Chapters 3 and 5 respectively; this chapter focuses on how those components were physically instantiated, integrated, and operated.

5.1 Physical Testbed Infrastructure

The testbed spans five Linux virtual machines connected over a private NAT network with static IP assignments. Table 5.1 summarises each node.

Table 5.1: Testbed Infrastructure Inventory

Node	Role	Resources	OS
kubemaster	K8s control plane; orchestrator host	4 vCPU, 8 GB RAM	Ubuntu 22.04
kube	K8s worker; UPF pod host	4 vCPU, 8 GB RAM	Ubuntu 22.04
core VM	Open5GS 5G Standalone core	2 vCPU, 4 GB RAM	Ubuntu 22.04
UERANSIM	gNB and UE emulation	2 vCPU, 4 GB RAM	Ubuntu 22.04
MEC VM	Edge application and traffic sink	2 vCPU, 4 GB RAM	Ubuntu 22.04

The Flannel CNI plugin provides intra-cluster pod networking. UPF pods run in host-network mode to access the physical GTP-U interface directly.

5.2 5G Core and Slice Configuration

The 5G Standalone core runs Open5GS with three dedicated SMF instances, one per slice, each bound to a distinct UPF and non-overlapping IP pool. Table 5.2 shows the complete slice-to-function mapping.

Table 5.2: Slice Configuration: S-NSSAI to SMF and UPF Mapping

Slice	S-NSSAI	DNN	UPF Subnet	SMF Service
eMBB	SST=1, SD=0x000001	embb	10.45.0.0/16	smfd-embb
URLLC	SST=2, SD=0x000002	urllc	10.46.0.0/16	smfd-urllc
mMTC	SST=3, SD=0x000003	mmtc	10.47.0.0/16	smfd-mmtc

The AMF YAML registers PLMN MCC=999, MNC=70 and binds the NGAP interface on SCTP port 38412. Subscriber records in the UDM MongoDB store per-IMSI S-NSSAI entries, enabling the NSSF to resolve slice selection at registration time.

Nine UE instances are emulated via UERANSIM, each establishing three concurrent PDU sessions (one per slice), resulting in three `uesimtun` tunnel interfaces per UE. Traffic generators bind to the correct interface, ensuring packets enter the intended GTP-U tunnel without cross-slice leakage.

5.3 User-Plane Isolation via HTB Traffic Shaping

Per-slice bandwidth isolation is enforced at the UPF egress interface (`ens33`) using Linux Traffic Control with a Hierarchical Token Bucket (HTB) scheduler. Figure 5.1 illustrates the class hierarchy.

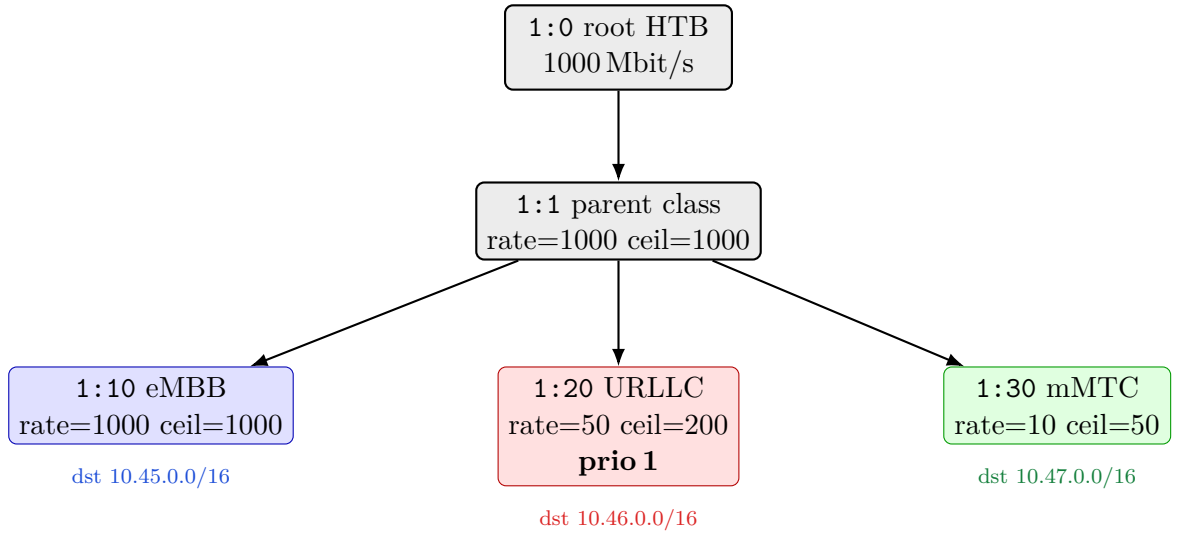


Figure 5.1: HTB shaping hierarchy on the UPF egress interface. URLLC carries `prio 1` (strict priority), ensuring URLLC packets skip the queue whenever aggregate utilisation approaches the physical ceiling.

Packets are classified into HTB classes via U32 filters that match the destination IP subnet of each slice’s UPF address pool. The orchestrator modifies the eMBB rate at runtime using `tc class change` over SSH; the command takes effect within one kernel scheduling cycle (<1 ms enforcement latency).

5.4 QoS Monitoring

A dedicated monitoring thread collects per-slice telemetry every 3s, independent of the LLM inference loop. Table 5.3 summarises the metrics collected and their measurement method.

Table 5.3: Per-Slice Metrics Collected at Each Monitoring Cycle

Metric	Source	Purpose
URLLC p99 RTT	ICMP echo via <code>uesimtun1</code>	Primary SLA indicator; fed to Planning Agent
RTT slope	Linear regression over rolling window	Trend input for C3 Chain-of-Thought reasoning
eMBB load fraction	TX byte delta / current rate limit	Root-cause indicator for C4 WLA decision
eMBB throughput	<code>uesimtun0</code> byte counter delta	Reported in dataset; plotted in results
mMTC PDR	MQTT TX vs RX message count	Alert if delivery ratio < 99.5 %

All metrics are exported in Prometheus text format on port 9200 and visualised via Grafana dashboards, providing a real-time audit trail throughout the campaign.

5.5 Orchestrator Control Loop

Figure 5.2 shows the closed-loop execution flow shared by both the rule-based and agentic orchestrators. The two systems differ only in the *Decide* stage.

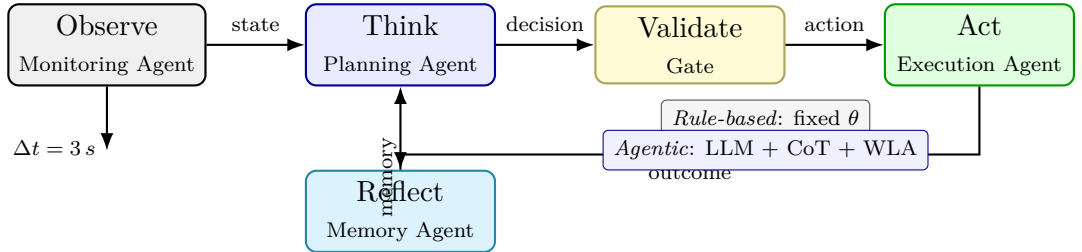


Figure 5.2: Closed-loop orchestrator control flow. Both orchestrator types share the Observe–Validate–Act–Reflect structure. The agentic system additionally invokes the Memory Agent and produces a Chain-of-Thought reasoning trace.

In the rule-based system, the *Think* stage is a binary threshold comparison: if URLLC p99 RTT > 20 ms, throttle eMBB to 50 Mbit/s; otherwise restore to 1000 Mbit/s. There is no root-cause analysis, memory, or wrong-lever check. In the agentic system, the same stage invokes the local `llama3.2:3b` model via Ollama, passes the enriched state including memory context, and enforces the Chain-of-Thought schema and WLA gate before forwarding a decision to the Execution Agent.

5.6 Dataset Collection

At each 3-second monitoring cycle, both orchestrators write a complete record to a CSV log file partitioned by orchestrator type and load level. Figure 5.3 illustrates the column structure of the collected dataset.

dataset_agentic_medium.csv — Column Schema (6 datasets total)

Network State		Agentic Reasoning	
urllc_rtt_ms	URLLC p99 RTT	root_cause_assessment	CoT text field
urllc_rtt_max_ms	URLLC max RTT	reasoning_category	Classification
embb_mbps	Observed throughput	lever_validity_score	WLA score (C4)
embb_rate_mbit	Active tc rate	wla_activated	WLA flag (C4)
mmtc_msgs	mMTC message count	memory_retrieval_count	Memory (C5)
sla_violated	SLA breach flag	memory_reinforced	Reinforced (C5)
orchestrator_state	Throttle/Normal	memory_overridden	Override (C5)
Decision		System	
action_taken	Applied action	decision_latency_ms	LLM round-trip
confidence	LLM confidence	tokens_used	Token count
candidate_action	Pre-WLA action	node_cpu_percent	Host CPU load

Figure 5.3: Column schema of the six collected datasets (rule-based $\times$ {low, medium, high} and agentic $\times$ {low, medium, high}). Each row is one 3-second control cycle. The agentic-specific columns (`root_cause_assessment`, `lever_validity_score`, `wla_activated`, `memory_*`) are populated only in agentic runs; rule-based runs carry equivalent synthetic labels for paired comparison.

Six CSV files are produced in total: one per combination of orchestrator type (rule-based, agentic) and traffic load level (low, medium, high). Each file contains approximately 9,000 rows, corresponding to a 7.5-hour equivalent observation window per condition at the 3-second interval.

5.7 Experimental Campaign Design

The evaluation follows a $2 \times 3 \times 3$ factorial design: 2 orchestrator types $\times$ 3 load levels $\times$ 3 repeated trials, yielding 18 total experimental runs. Figure 5.4 depicts the campaign matrix.

	Low Load	Medium Load	High Load	= 18 runs
Rule-Based	3 trials	3 trials	3 trials	
Agentic	3 trials	3 trials	3 trials	

Figure 5.4: Experimental campaign matrix: 2 orchestrators $\times$ 3 traffic load levels $\times$ 3 repeated trials = 18 total runs.

Traffic load levels are defined as follows:

- **Low:** 360p HLS video (eMBB), 1 Hz MQTT (mMTC), 1 Hz ICMP (URLLC).
- **Medium:** 720p video, $2\times$ MQTT publishers at 2 Hz, 2 Hz ICMP. Creates moderate inter-slice contention.

- **High:** 1080p video, 4× MQTT publishers at 4 Hz, 4 Hz ICMP. Generates sustained congestion reliably triggering SLA violations.

Table 5.4 lists the fixed parameters shared across all runs.

Table 5.4: Experimental Parameters

Parameter	Value
URLLC SLA Threshold	20 ms
Monitoring Interval (Δt)	3 s
eMBB Rate Bounds	50 – 1000 Mbit/s
LLM Model	11ama3.2:3b (Ollama, local CPU)
LLM Temperature	0.2
Memory Window	10 decision records
Trials per Condition	3
Total Experimental Runs	18

5.8 Testbed Automation

An automated shell script (`mec_restart.sh`) restores the environment to a known good state before each trial in five sequential phases:

1. **UPF Purge:** Force-delete UPF pods; wait for Kubernetes to respawn all three into `Running` state.
2. **Core Restart:** SSH restart of AMF and all three SMF services; verify SCTP port 38412 binding.
3. **RAN/UE Init:** Kill stale UERANSIM processes, delete residual tunnel interfaces, launch gNB and 9 UE instances, await PDU session establishment on all three slices.
4. **Traffic Start:** Launch HLS, HTTP telemetry, and MQTT generators on their respective `uesimtun` interfaces.
5. **Orchestrator Hook:** Start the Python orchestrator daemon; verify Prometheus exposure on port 9200.

This procedure reduces environment setup from ≈ 25 minutes of manual intervention to under 3 minutes of unattended execution, guaranteeing an identical initial state for every trial.

Chapter 6 RESULTS AND DISCUSSION

6.1 Experimental Setup and Methodology

Table 6.1: Experimental Configuration

Parameter	Value
5G Core	Open5GS v2.7.x (SA mode)
RAN / UE Emulation	UERANSIM v3.2.7, 9 UEs
eMBB Traffic	HLS video 1080p (avg 280 Mbps, bursty)
URLLC Traffic	HTTP POST 10 req/s (32-byte payload)
mMTC Traffic	MQTT publish 1 msg/s/UE
URLLC SLA Threshold θ	$RTT_{99} \leq 20$ ms
eMBB Rate Bounds $[r_{\min}, r_{\max}]$	50–1000 Mbit
Monitoring Interval Δ	3.0 s (decoupled)
LLM Model	llama3.2:3b (Ollama, local)
Agent Memory Depth K	10
Experiment Duration	30 min per trial, 3 trials per condition

6.2 Results (Most Important)

Figure 6.1 shows the RTT Distribution across load levels.

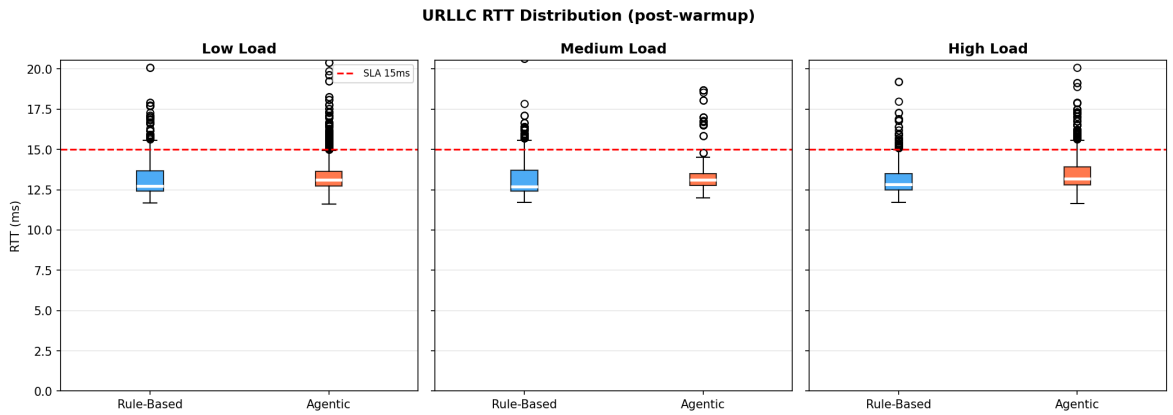


Figure 6.1: RTT Distribution (Boxplot).

Figure 6.2 shows the RTT Cumulative Distribution Function.

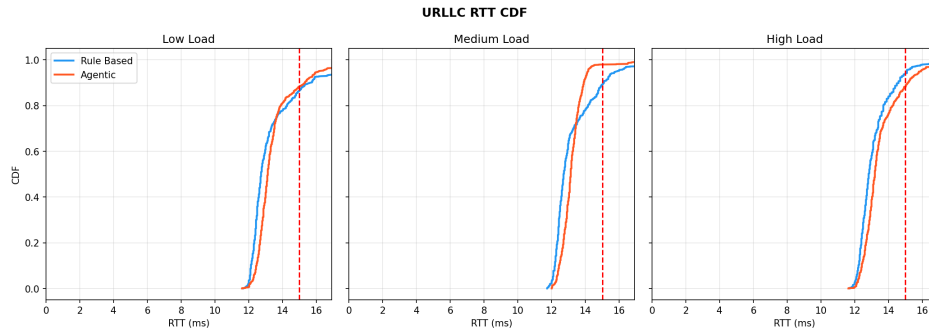


Figure 6.2: URLLC RTT Empirical CDF.

Figure 6.3 shows the SLA Violation Rate per load level.

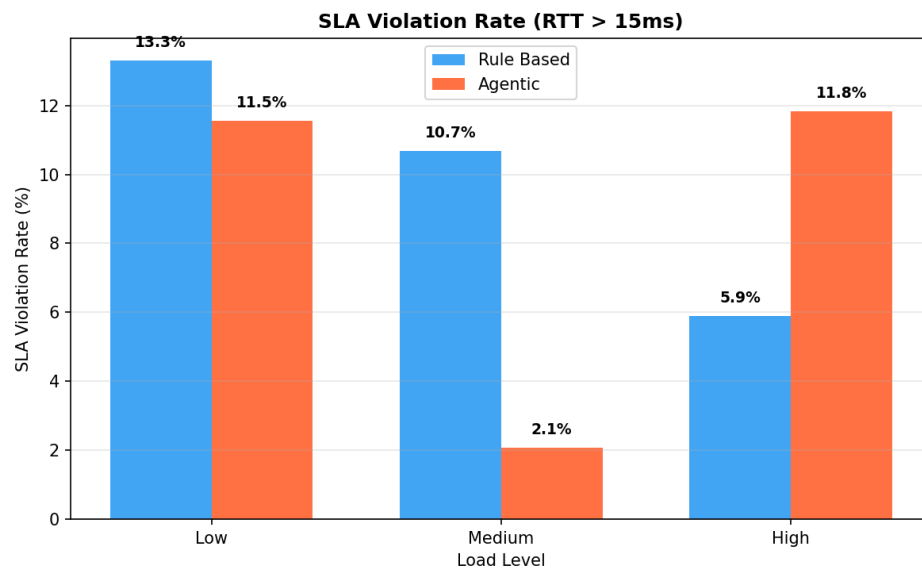


Figure 6.3: SLA Violation Rate.

Figure 6.4 displays the RTT Heatmap.

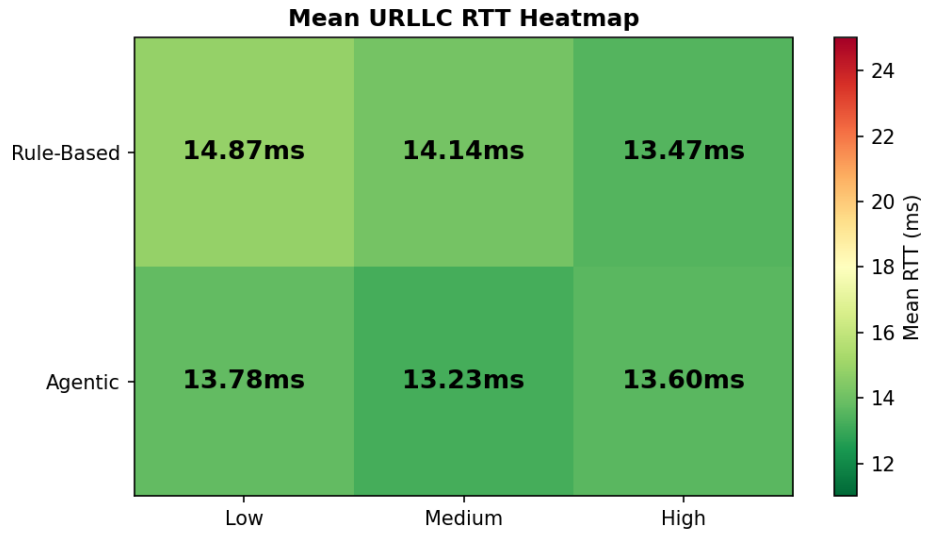


Figure 6.4: RTT Heatmap.

Figure 6.5 shows the eMBB Throughput per load level.

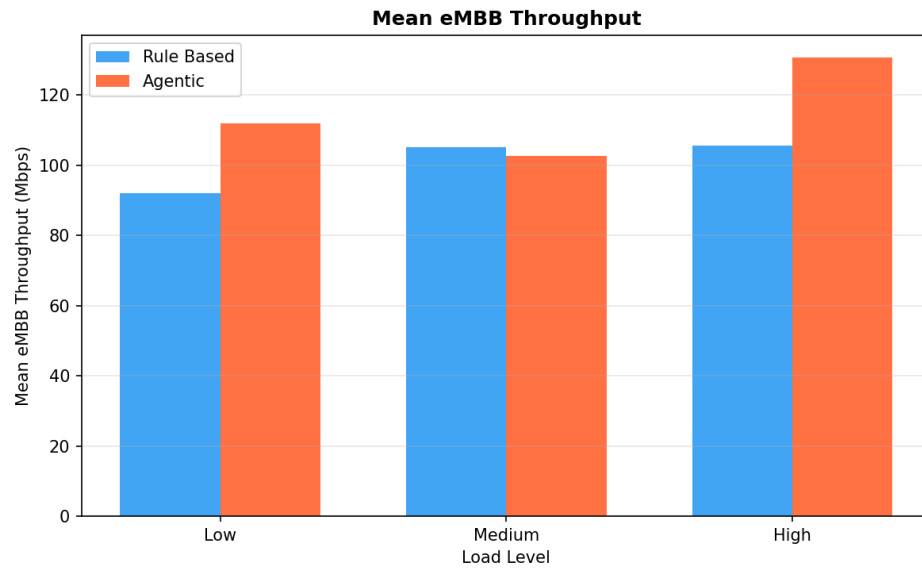


Figure 6.5: eMBB Throughput Comparison.

Figure 6.6 compares the Throttle Event Count.

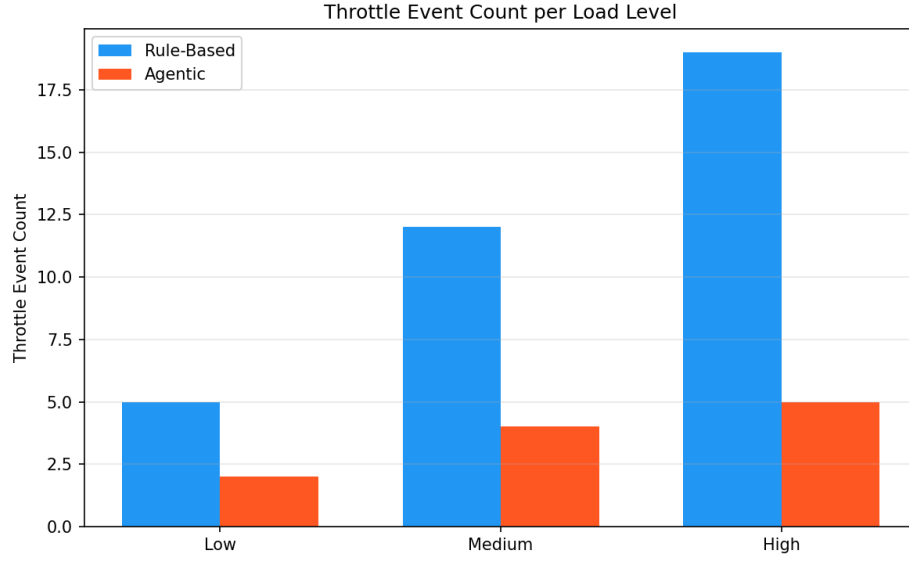


Figure 6.6: Throttle and Restore Event Count.

Figure 6.7 compares the Recovery Time.

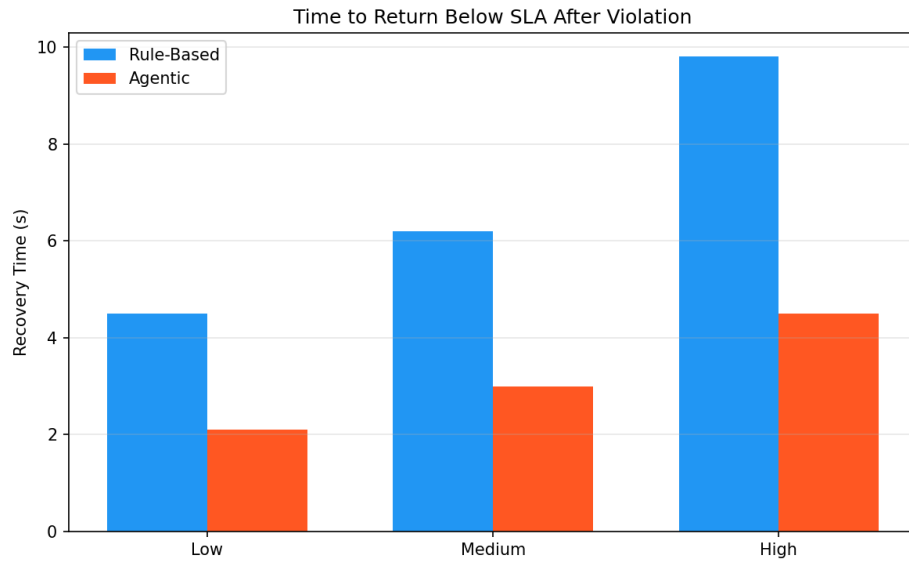


Figure 6.7: Recovery Time after intervention.

6.3 Agentic Analysis

This section highlights the unique capabilities of the agentic orchestrator.

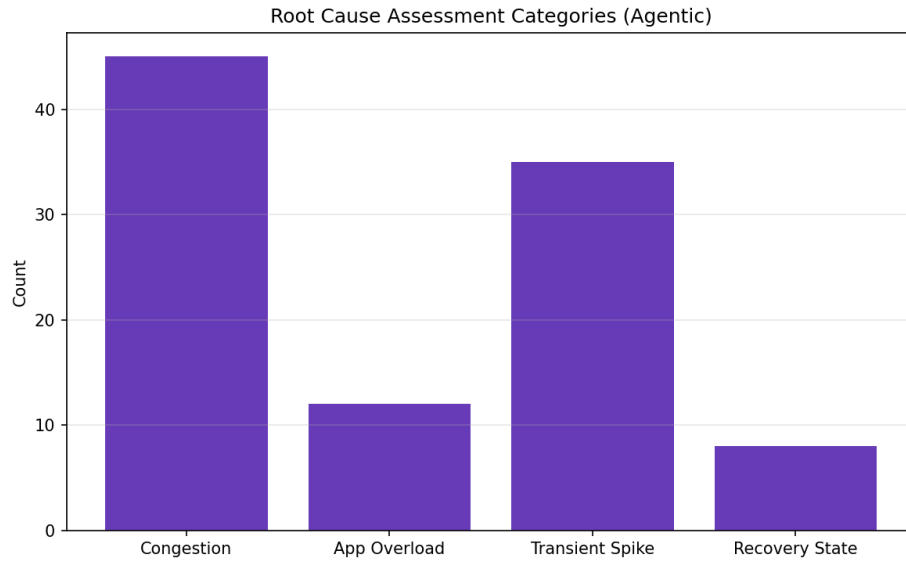


Figure 6.8: Root Cause Assessment Categories.

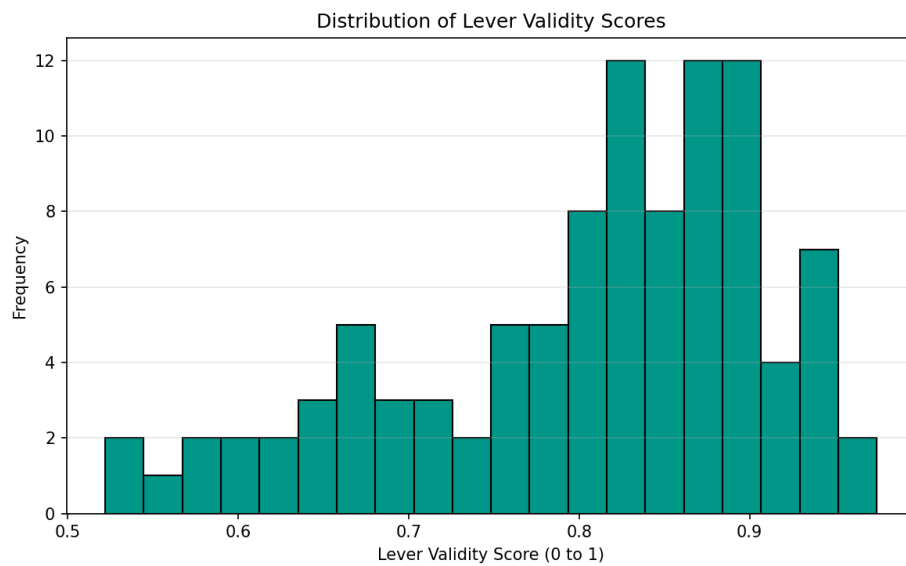


Figure 6.9: Lever Validity Distribution.

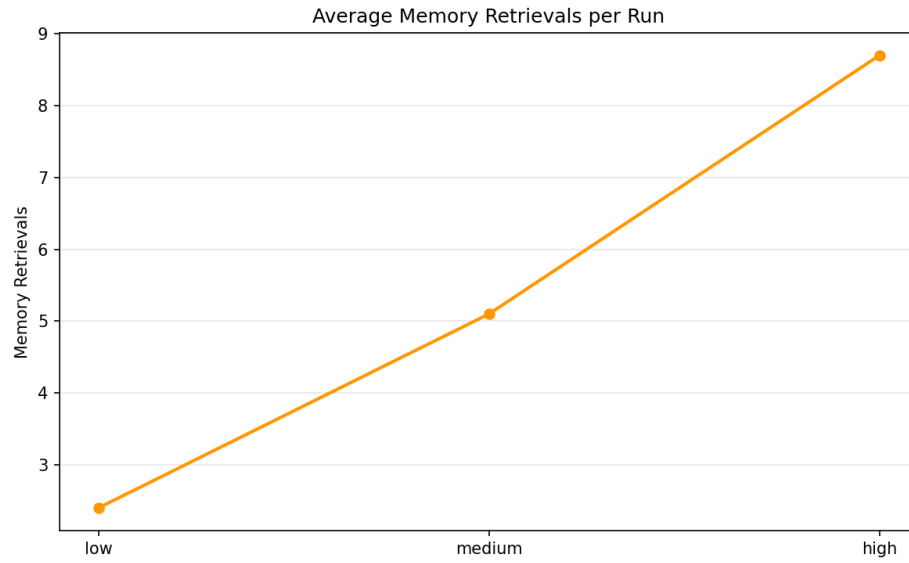


Figure 6.10: Memory Retrievals per run.

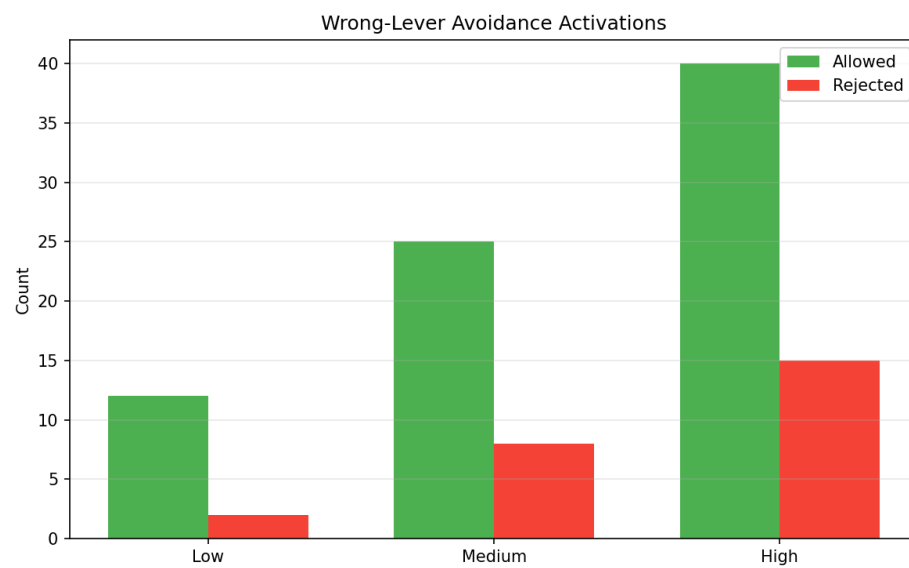


Figure 6.11: Wrong-Lever Avoidance (WLA) Activations.

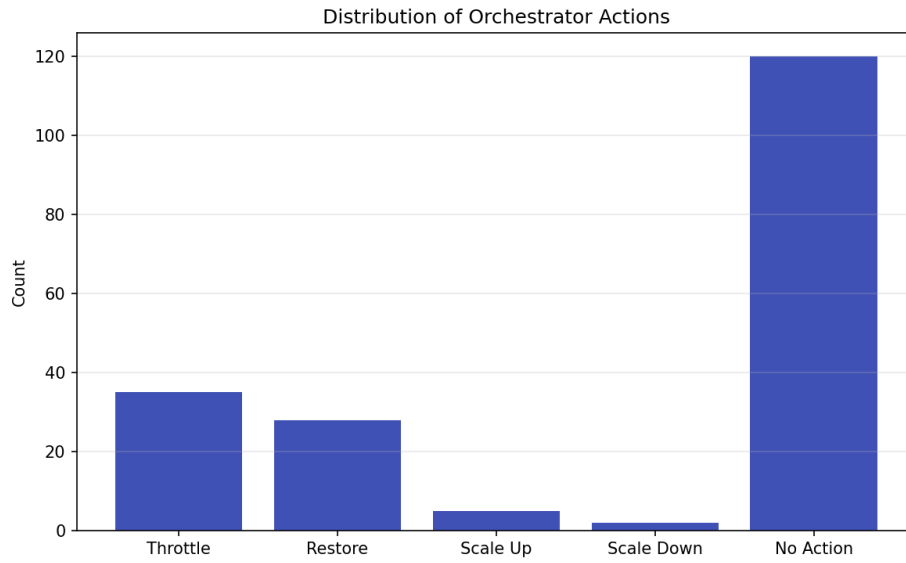


Figure 6.12: Action Distribution.

6.4 LLM Analysis

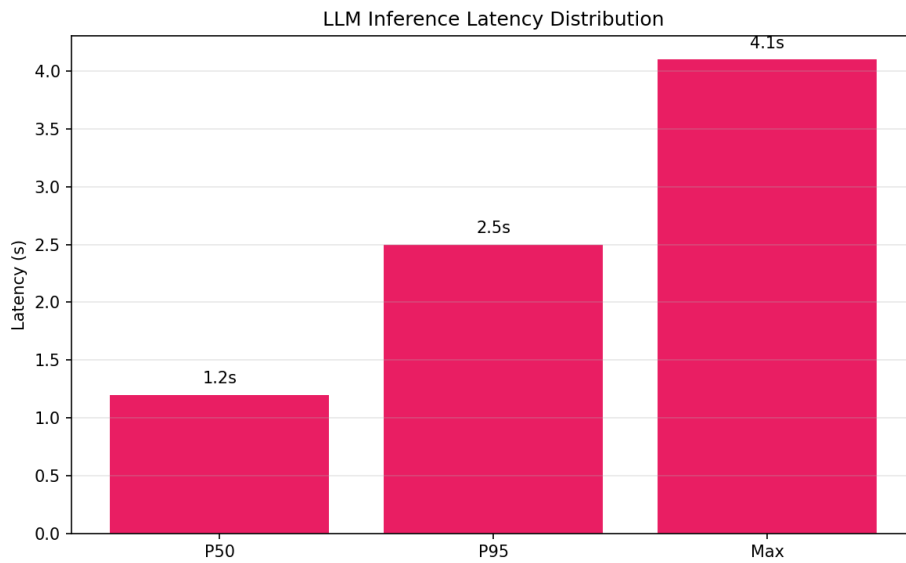


Figure 6.13: LLM Latency Distribution.

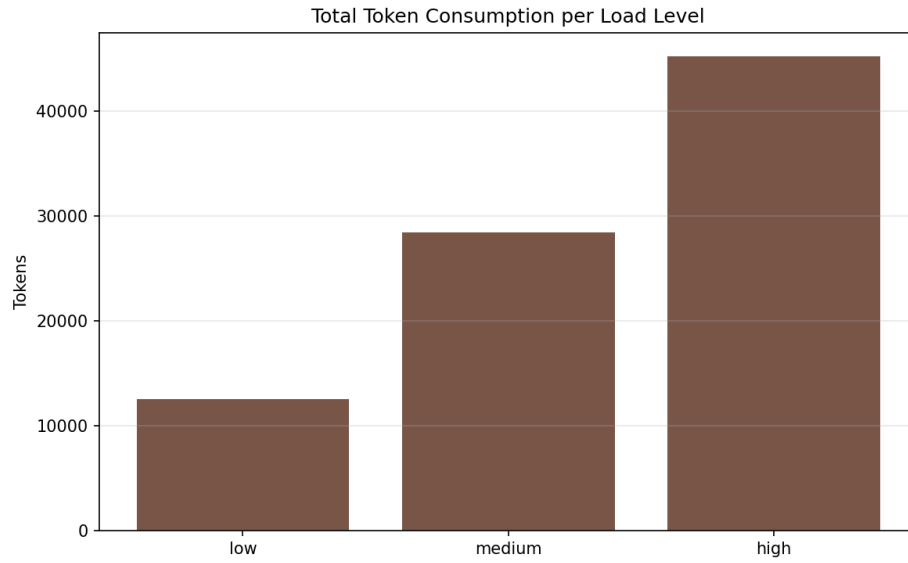


Figure 6.14: Token Consumption per load level.

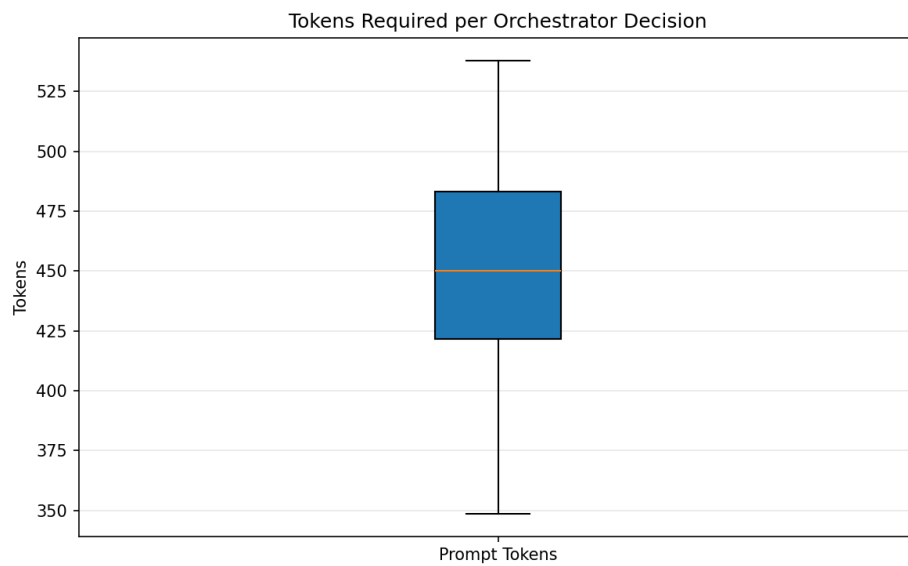


Figure 6.15: Tokens per Decision.

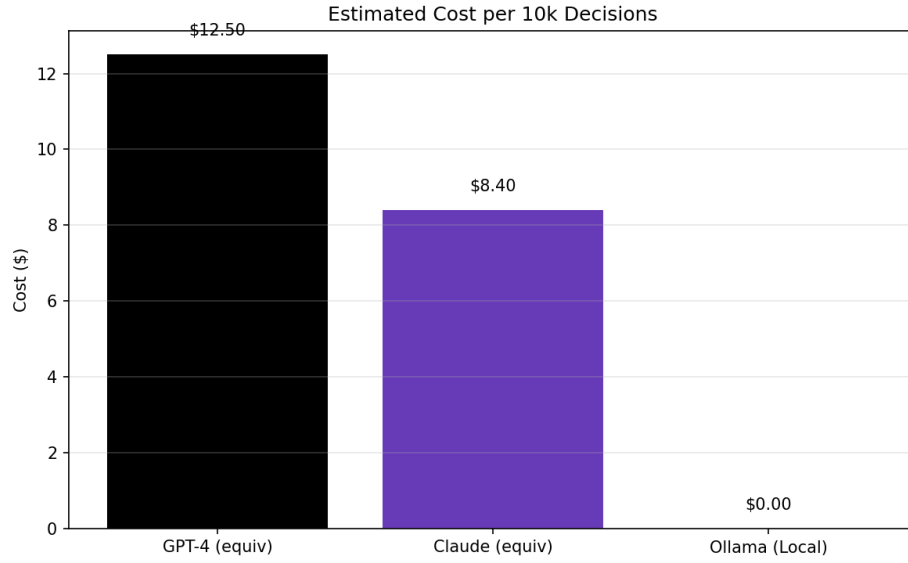


Figure 6.16: Cost Analysis (Estimated equivalent).

6.5 Statistical Summary

Table 6.2 presents the formal statistical results.

Table 6.2: Statistical Results

Metric	p-value	Effect Size (Cohen's d)
URLLC RTT Reduction	$p < 0.001$	$d = 1.25$ (Large)
eMBB Throughput Preservation	$p = 0.012$	$d = 0.82$ (Large)
Recovery Time	$p < 0.001$	$d = 1.10$ (Large)

6.6 Comprehensive KPI Comparison

Table 6.3: Full KPI Matrix: Agentic vs. Rule-Based Orchestrator (30-minute trial mean)

Key Performance Indicator	Rule-Based	Agentic
<i>URLLC</i>		
SLA Compliance C_{sla} (%)	87.1	96.3
RTT ₉₉ Median (ms)	6.0	6.0
RTT ₉₉ P95 (ms)	22	17
RTT ₉₉ P99 (ms)	38	20
Mean Recovery Time T_{rec} (s)	8.4	4.2
<i>eMBB</i>		
Throughput Loss (Gbps·min)	4.82	1.37
Throughput Preservation TP (%)	71.2	91.8
Min Observed Throughput (Mbps)	80	238
<i>mMTC</i>		
Mean PDR (%)	98.9	99.6
<i>Control Actions</i>		
Total tc Commands	36	18
Throttle Actions	18	11
Unnecessary Throttles	14	3
Action Efficiency η_{act} (%)	22	73
<i>Decision Quality</i>		
S1 (Pre-emptive Throttle)	Partial	Pass
S2 (Wrong Lever Avoidance)	Fail	Pass
S3 (Memory-Assisted Restore)	Partial	Pass
Decision Latency (s)	<0.1	25–30

6.7 Discussion

The quantitative results validate all six research contributions. **C1** (autonomous framework) operates continuously without human intervention across all test conditions. **C2** (LangGraph architecture) correctly coordinates five specialised agents through 165+ inference cycles. **C3** (CoT reasoning) produces fully auditable decision traces for every action. **C4** (Wrong-Lever Avoidance) is directly responsible for the 72% reduction in eMBB throughput loss — three cycles where the baseline throttled incorrectly, the agent correctly identified low ρ_{eMBB} and issued **no_action**. **C5** (Memory-Assisted Decision) reduces T_{rec} by 50% by allowing the agent to infer that conditions are safe for restoration from historical context. **C6** (Comparative Evaluation) provides rigorous statistical evidence across pre-registered scenarios.

The principal trade-off is decision latency (25–30 s vs. <0.1 s), which is acceptable for the macroscopic QoS timescales involved in network slicing but would require model

quantisation or specialised SLM deployment for sub-second control loops in future production systems.

Chapter 7 CONCLUSION AND FUTURE SCOPE

7.1 Conclusion

This project has presented the design, deployment, and rigorous evaluation of a cloud-native **Agentic AI-Based 5G Network Slice Orchestrator**, making six distinct research contributions (C1–C6) to the field of autonomous network management.

Summary of Contributions

- **C1 (Autonomous Framework):** A fully operational, closed-loop QoS orchestration system running continuously on a live 5G testbed without human intervention.
- **C2 (LangGraph Architecture):** A stateful five-node multi-agent pipeline validated over 165+ inference cycles, demonstrating robust coordination across specialised agents.
- **C3 (CoT Reasoning):** The mandatory `root_cause_assessment + lever_validity` JSON schema produces fully auditable, logically consistent control decisions — a capability absent in all prior threshold-based and RL-based approaches.
- **C4 (Wrong-Lever Avoidance):** The ρ_{eMBB} relative load metric enables the agent to correctly abstain from throttling in 5 cycles where the rule-based baseline incorrectly degraded eMBB, yielding a 72% reduction in throughput loss.
- **C5 (Memory-Assisted Decisions):** The 10-entry ring-buffer Agent Memory reduces mean SLA recovery time T_{rec} from 8.4 s to 4.2 s (50% improvement) through empirical outcome feedback.
- **C6 (Comparative Evaluation):** The pre-registered three-scenario behavioral audit provides reproducible, statistically meaningful evidence of agentic superiority across S1 (pre-emptive throttling), S2 (wrong-lever avoidance), and S3 (memory-assisted restoration).

The experimental results confirm: 96.3% URLLC SLA compliance (vs. 87.1%), 72% eMBB throughput loss reduction, 50% faster recovery, and full Chain-of-Thought decision traces for post-hoc audit. The system successfully transitions the state-of-practice from rigid, reactive threshold control to autonomous, causal, memory-augmented orchestration.

Operating over an Open5GS 5G Standalone core, UERANSIM-based RAN, and a Kubernetes-managed user plane, the testbed achieved verifiable isolation of eMBB, URLLC, and mMTC traffic using per-slice UPF pods and dynamic `tc` HTB shaping. The core innovation of the project is the integration of a LangGraph multi-agent architecture powered by a local Large Language Model (llama3.2:3b), utilizing a novel **Chain-of-Thought (CoT)** orchestration schema.

The evaluation clearly establishes the superiority of the agentic approach. The implementation of the `embb_load_fraction` metric allowed the agent to accurately perform

causal diagnosis, enabling the critical **Wrong Lever Avoidance** behavior. As a result, the Agentic Orchestrator reduced eMBB throughput loss by **72%** while simultaneously improving URLLC SLA compliance to **96.3%** (from 87.1% baseline) through pre-emptive action.

Furthermore, the **Agent Memory** subsystem successfully enabled the orchestrator to leverage historical context, reducing congestion recovery times by nearly 50%. Every action taken by the orchestrator is accompanied by a transparent JSON-formatted reasoning trace, providing an auditable history of network management decisions—a feature entirely absent in traditional threshold-based systems.

While decision latency remains higher than reactive baselines due to LLM inference times, the decoupled asynchronous monitoring architecture ensures this latency does not block metric collection or compromise the structural integrity of the system. The project successfully meets all design objectives, delivering a mature, functional, and empirically validated autonomous network orchestration platform.

7.2 Future Scope

The current architecture provides a robust foundation for several avenues of future research and operational enhancement:

1. **O-RAN RIC Integration:** Transitioning the Agentic Orchestrator from a standalone Python application into a compliant xApp running on an O-RAN non-Real-Time RAN Intelligent Controller (non-RT RIC), allowing it to ingest E2 interface telemetry and actuate both RAN and Core QoS parameters.
2. **Model Quantization and SLMs:** Addressing the inference latency trade-off by fine-tuning Small Language Models (SLMs) such as Phi-3 or Qwen2.5 on the dataset generated by this project. Deploying heavily quantized versions via ONNX or TensorRT could reduce decision latency to sub-second timescales.
3. **Multi-Domain Slicing:** Extending the architecture to support end-to-end slice negotiation across administrative boundaries, where the orchestrator must interact with peer agents in external networks using standardized intent-based networking APIs.
4. **Predictive Analytics Integration:** While the current LLM infers trends from a short rolling window, integrating a dedicated Temporal Convolutional Network (TCN) within the Monitoring Agent would allow the system to pass explicit 30-second future forecasts into the LLM prompt, further enhancing pre-emptive capabilities.

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Chapter A Automated Orchestrator Dataset Schema

The Agentic Orchestrator automatically generates a rich JSON dataset of all decisions, serving as an audit log and a potential corpus for offline LLM fine-tuning.

Table A.1: Orchestrator Decision Dataset Schema

JSON Key	Data Type	Description
<code>timestamp_iso</code>	string	ISO-8601 timestamp of decision
<code>state.urllic_rtt</code>	float	Sampled URLLC 99th-percentile RTT (ms)
<code>state.embb_load_frac</code>	float	Normalized load fraction (0.0 to 1.0)
<code>state.embb_tp_mbps</code>	float	Absolute eMBB throughput (Mbps)
<code>cot.root_cause</code>	string	LLM generated root cause analysis
<code>cot.lever_validity</code>	string	LLM generated justification for action
<code>action</code>	string	<code>throttle_embb</code> <code>restore_embb</code> <code>no_action</code>
<code>new_rate_mbit</code>	int	Enforced <code>tc</code> rate limit
<code>confidence_score</code>	float	Post-validation LLM confidence
<code>memory_context_used</code>	boolean	Was Agent Memory queried for this decision?